UNIVERSITY OF RWANDA
COLLEGE OF SCIENCE & TECHNOLOGY
SCHOOL OF ICT
DEPARTMENT OF COMPUTER ENGINEERING
MODULE :
REG NO: 223007703

Lab 6: Push Notifications with FCM.

A. I started by creating a new Flutter project and adding the firebase_core, firebase_messaging, and flutter local notification packages to pubspec.yaml. I then went to the Firebase console, created a new project and registered my Android app using the package com.company.fcml. I downloaded the google-services.json file and place it inside android/app/.
I ran the flutter fire CLI command "Flutterfire configure" to generate the firebase_options.dart file with my project credentials.

In main.dart, I initialized Firebase, registered a background message handler, requested notification permission at runtime and retrieved the FCM device token. I displayed the token on the screen, then opened Firebase console, went to cloud messaging, created a test message, pasted the token and sent it.

B. When the app was open (foreground), the notification arrived immediately and a popup dialog appeared on screen with the title and body of the message. The notification was also added to the list displayed inside the app UI.

CamScanner

When the app was in the background, the notification appeared in the phone status bar like a normal system notification. Tapping it opened the app. When I killed the app completely and sent a notification, it still appeared in the notification bar, and when I tapped it the app launched and showed the message content. The FCM token was unique to my device.

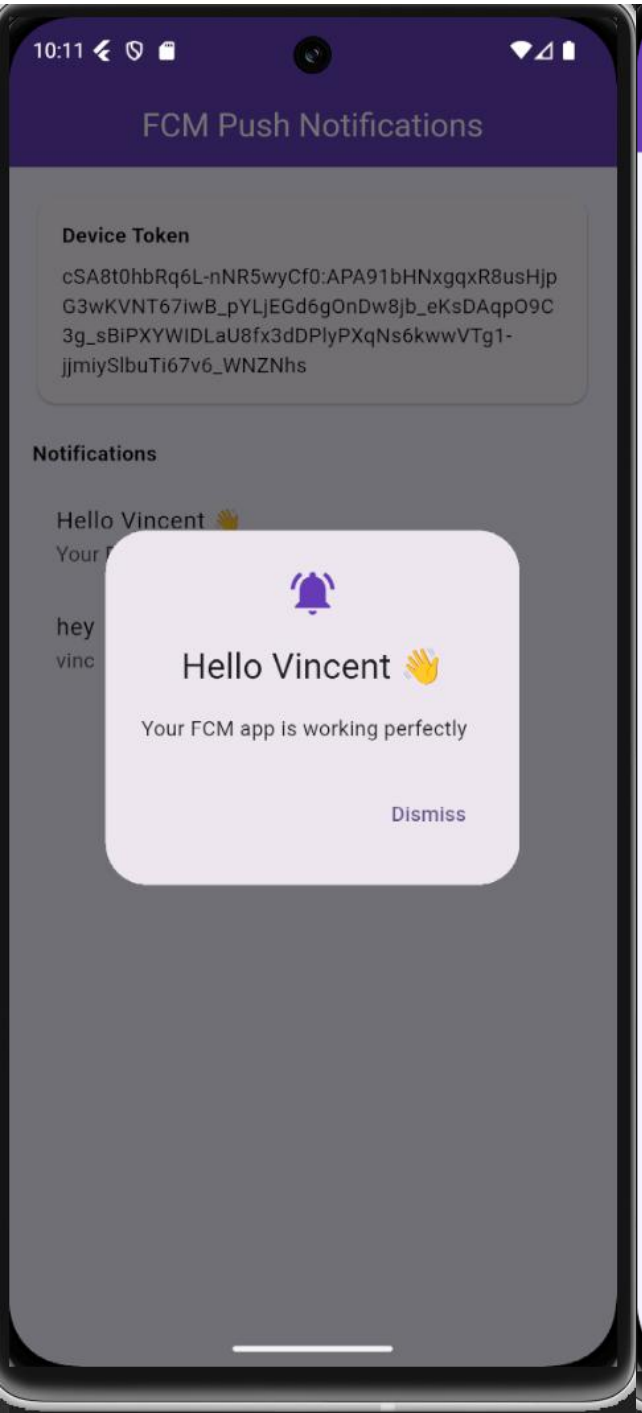
C. At first the app crashed on launch because the google-services.json file was missing. I fixed this by downloading the correct file from Firebase console and placing it in the android/app/ directory. I also got a Gradle build error because minSDK version was set too low. I changed it to 21 in the app build.gradle file and the build succeeded. On Android 13, the notification permission dialog did not appear automatically. I had to call requestPermission() explicitly in the code to trigger it.

Background messages did not work until I annotated the handler function with @pragma and made it a top-level function.

D. I learned how Firebase Cloud Messaging works: When a message is sent from the Firebase console FCM routes it to the target device using its unique registration token. The device token identifies a specific app installation.

I understood the different b/w foreground and background notification handling in flutter. Foreground requires listening to onMessage stream, while background requires a separate top-level handler function.

I also learned that local notification are needed to show heads up banners when the app is in the foreground, since FCM alone does not display them. overall this lab showed me how real-time push notification work in mobile apps.


10:11
FCM Push Notifications
Device Token
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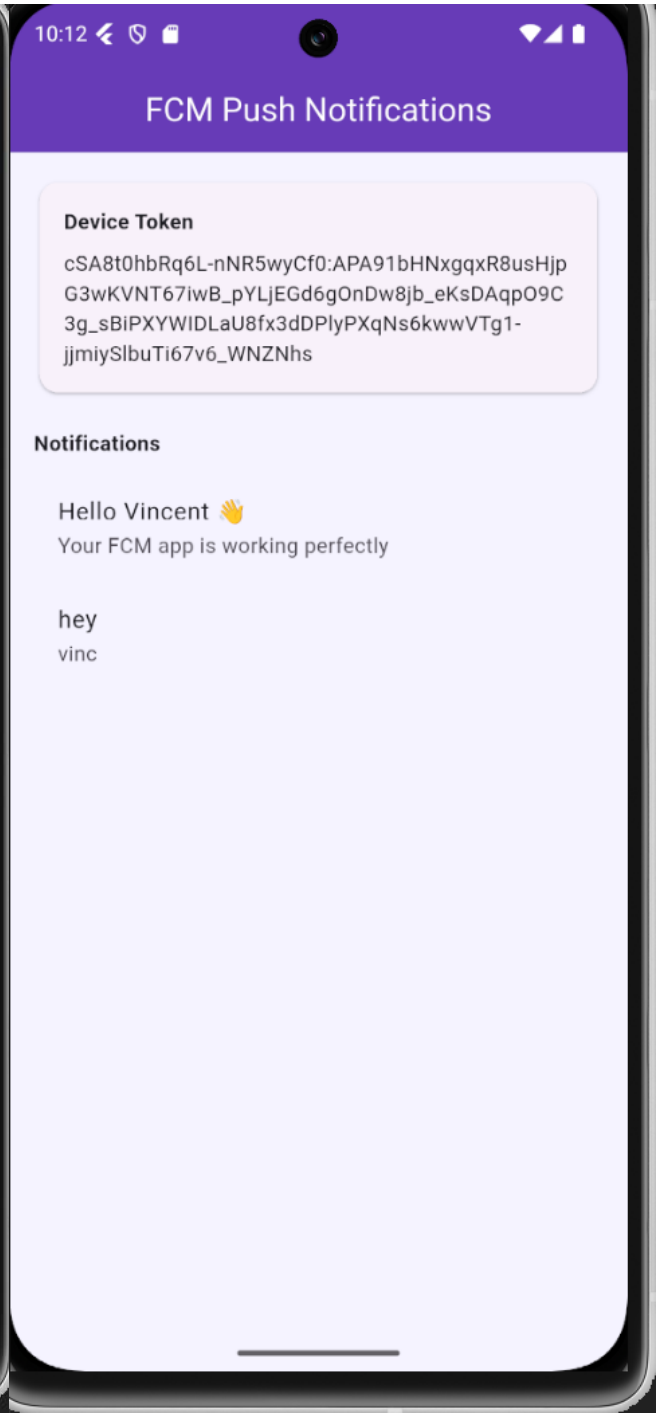
G3wKVNT67iwB_pYLjEGd6gOnDw8jb_eKsDAqpO9C
3g_sBiPXYWIDLaU8fx3dDPlyPXqNs6kwwVTg1-
jjmiySlbuTi67v6_WNZNhs
Notifications
Hello Vincent 👋
hey
vinc
Hello Vincent 👋
Your FCM app is working perfectly
Dismiss


10:12
FCM Push Notifications
Device Token
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3g_sBiPXYWIDLaU8fx3dDPlyPXqNs6kwwVTg1-
jjmiySlbuTi67v6_WNZNhs
Notifications
Hello Vincent 👋
Your FCM app is working perfectly
hey
vinc

1 Notification

Notification title

⚠ Urgent Update

Notification text

Please check your app immediately. Important info just arrived!

Notification image (optional)

Example: https://yourapp.com/image.png

Notification name (optional)

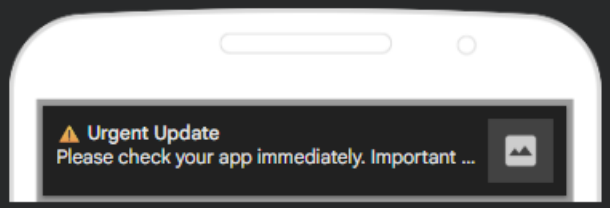
Enter optional name

Device preview

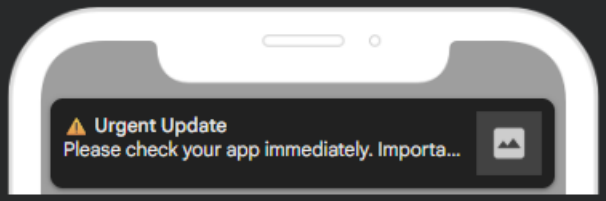
This preview provides a general idea of how your message will appear on a mobile device. Actual message rendering will vary depending on the device. Test with a real device for actual results.

Send test message

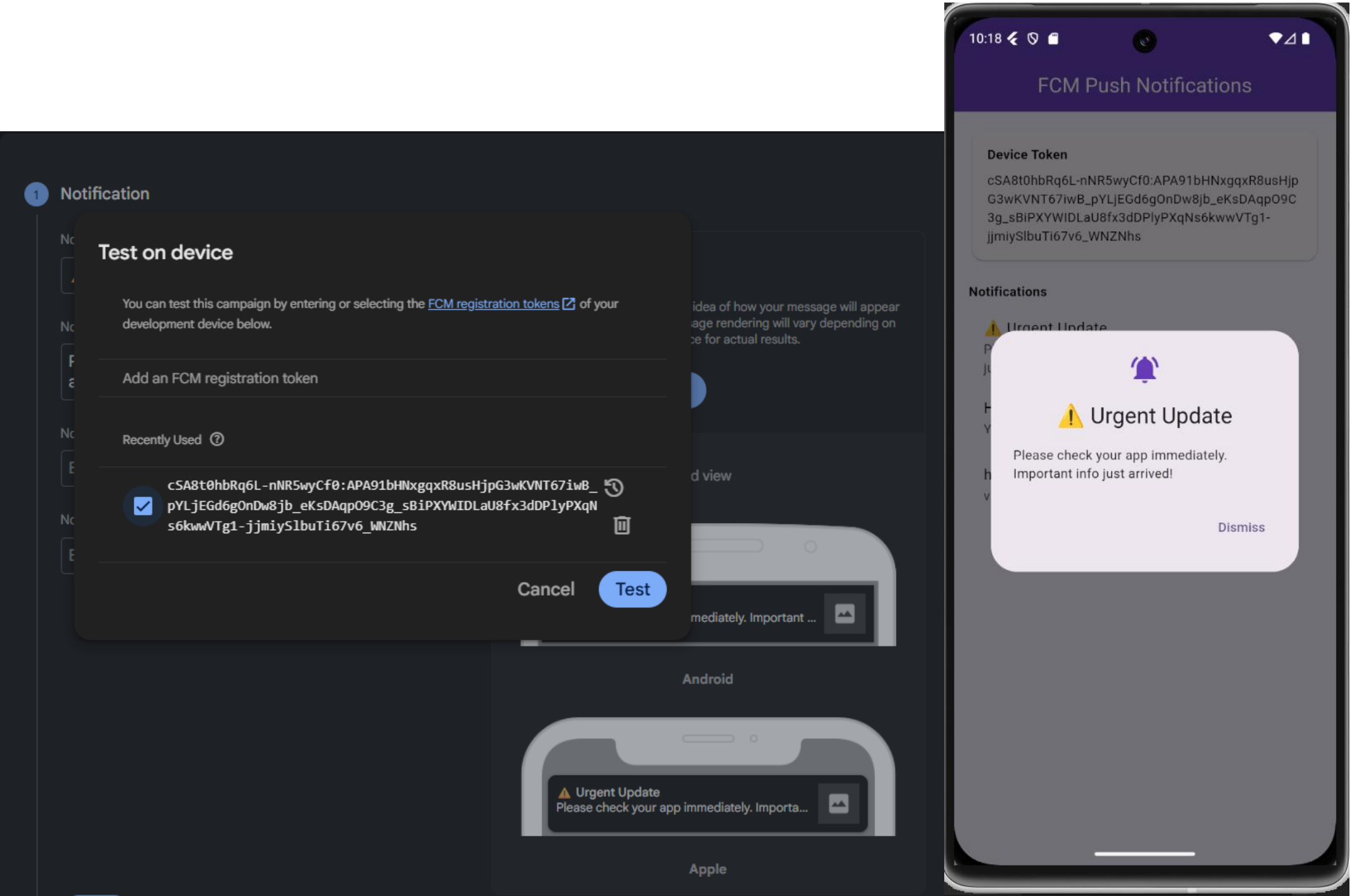
Initial state Expanded view

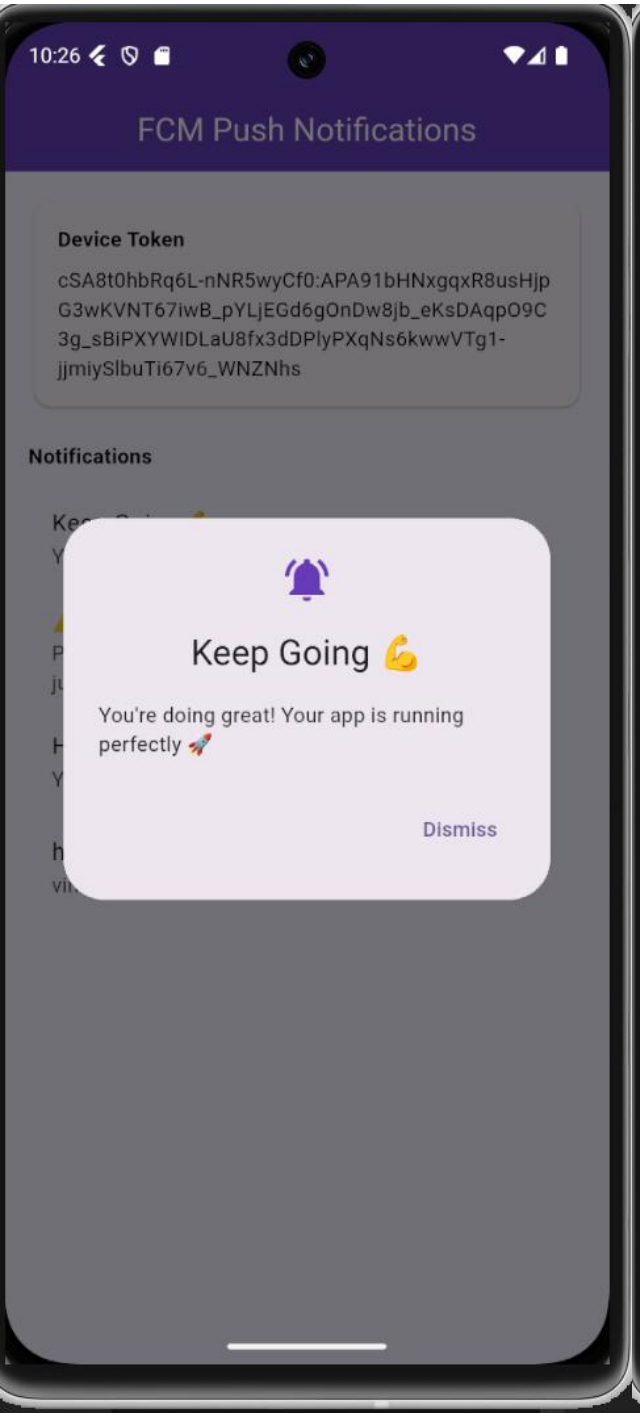


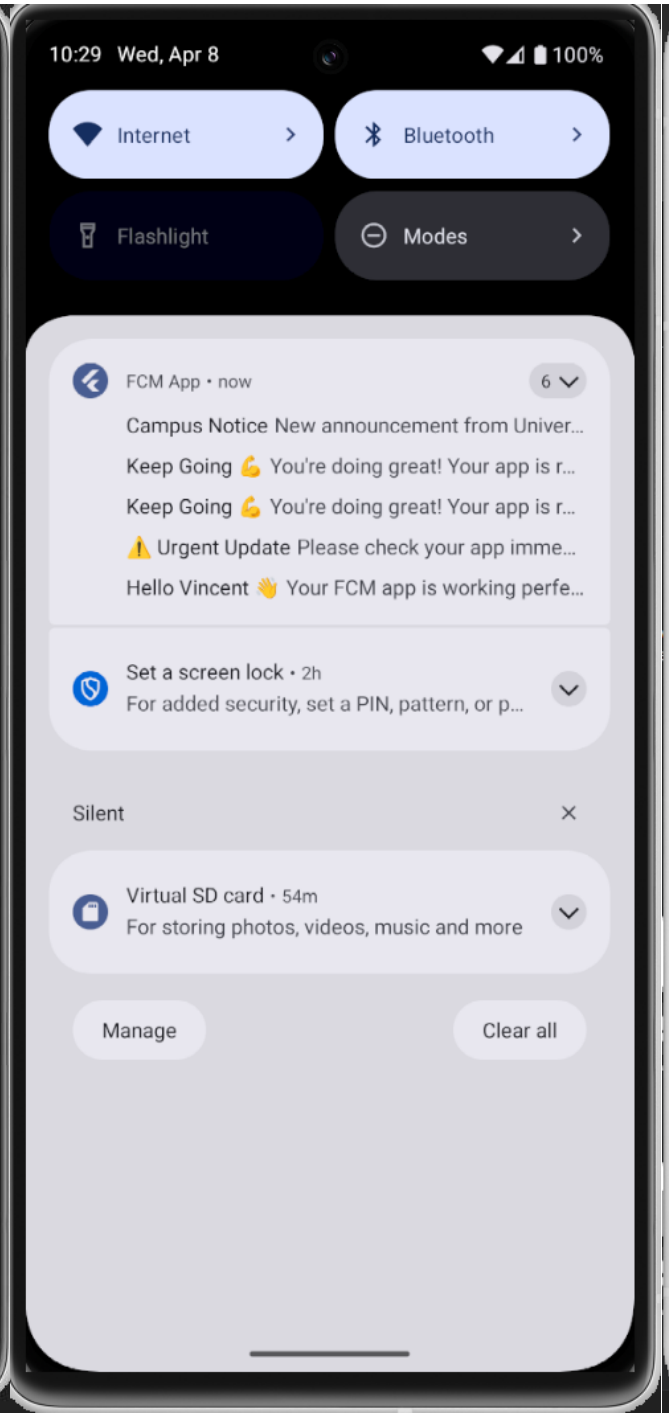
Android

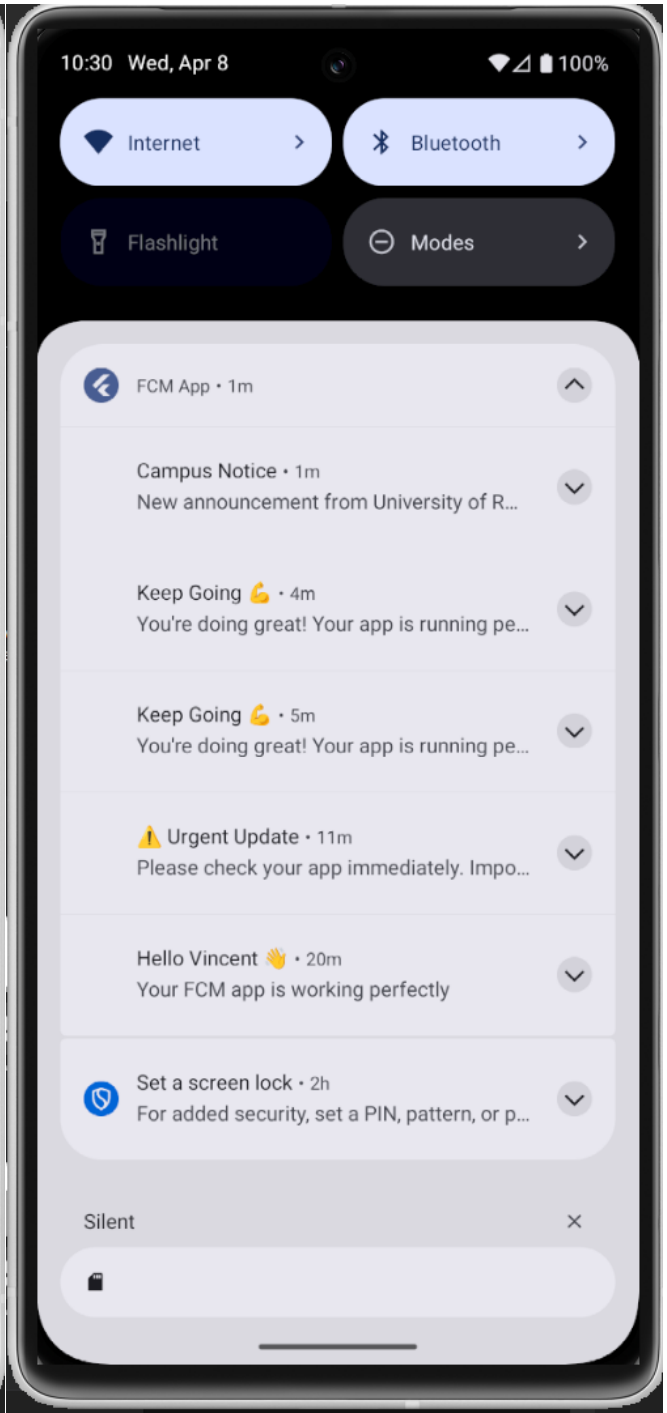


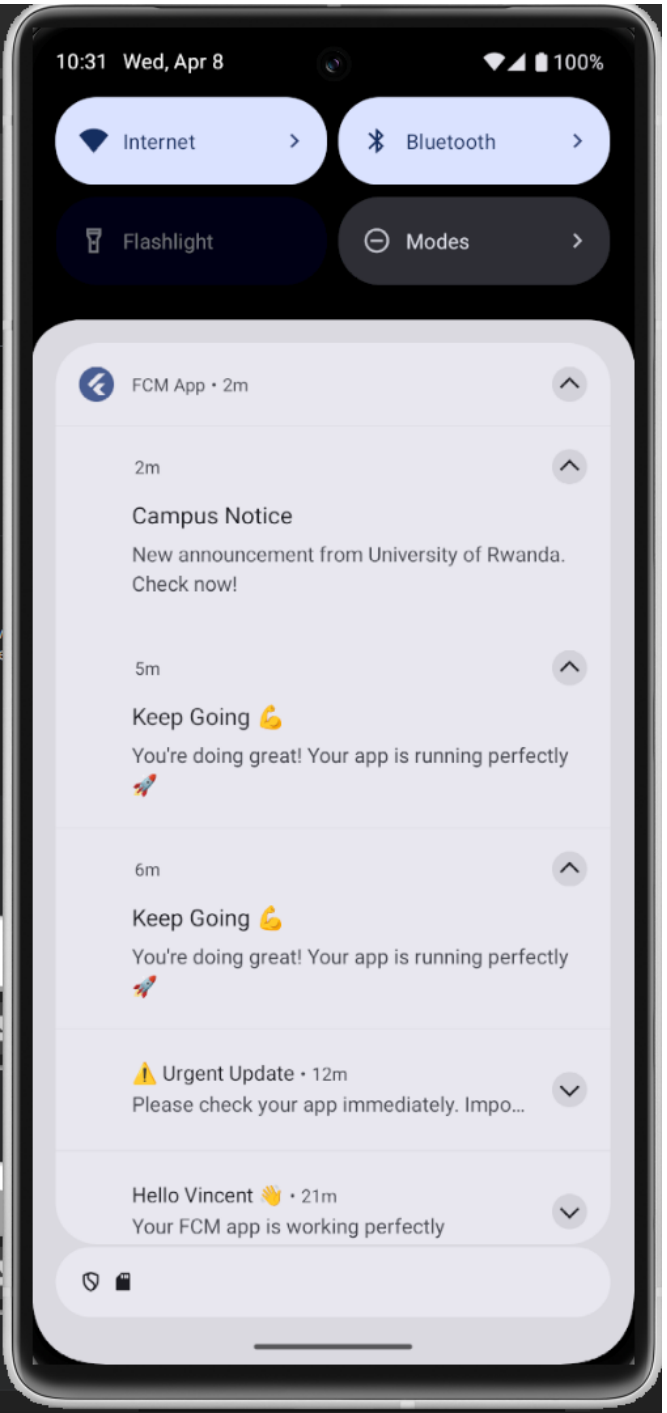
Apple


1 Notification
Test on device
You can test this campaign by entering or selecting the FCM registration tokens of your development device below.
Add an FCM registration token
Recently Used
cSA8t0hbRq6L-nNR5wyCf0:APA91bHNxgqxR8usHjpG3wKVNT67iwB_pYLjEGd6gOnDw8jb_eKsDAqpO9C3g_sBiPXYWIDLaU8fx3dDPlyPXqNs6kwwVTg1-jjmiySlbuTi67v6_WNZNhs
Cancel
Test
idea of how your message will appear
age rendering will vary depending on
ce for actual results.
d view
mediately. Important ...
Android
Urgent Update
Please check your app immediately. Importa...
Apple
10:18
FCM Push Notifications
Device Token
cSA8t0hbRq6L-nNR5wyCf0:APA91bHNxgqxR8usHjpG3wKVNT67iwB_pYLjEGd6gOnDw8jb_eKsDAqpO9C3g_sBiPXYWIDLaU8fx3dDPlyPXqNs6kwwVTg1-jjmiySlbuTi67v6_WNZNhs
Notifications
Urgent Update
Please check your app immediately.
Important info just arrived!
Dismiss


10:26
FCM Push Notifications
Device Token
cSA8t0hbRq6L-nNR5wyCf0:APA91bHNxgqxR8usHjpG3wKVNT67iwB_pYLjEGd6gOnDw8jb_eKsDAqpO9C3g_sBiPXYWIDLaU8fx3dDPlyPXqNs6kwwVTg1-jjmiySlbuTi67v6_WNZNhs
Notifications
Keep Going 💪
You're doing great! Your app is running perfectly 🚀
Dismiss


10:29 Wed, Apr 8
100%
Internet
Bluetooth
Flashlight
Modes
FCM App • now
6
Campus Notice New announcement from Univer...
Keep Going 💪 You're doing great! Your app is r...
Keep Going 💪 You're doing great! Your app is r...
⚠️ Urgent Update Please check your app imme...
Hello Vincent 👋 Your FCM app is working perfe...
Set a screen lock • 2h
For added security, set a PIN, pattern, or p...
Silent
Virtual SD card • 54m
For storing photos, videos, music and more
Manage
Clear all


10:30 Wed, Apr 8
100%
Internet
Bluetooth
Flashlight
Modes
FCM App • 1m
Campus Notice • 1m
New announcement from University of R...
Keep Going 💪 • 4m
You're doing great! Your app is running pe...
Keep Going 💪 • 5m
You're doing great! Your app is running pe...
⚠️ Urgent Update • 11m
Please check your app immediately. Impo...
Hello Vincent 👋 • 20m
Your FCM app is working perfectly
Set a screen lock • 2h
For added security, set a PIN, pattern, or p...
Silent


10:31 Wed, Apr 8
100%
Internet
Bluetooth
Flashlight
Modes
FCM App • 2m
2m
Campus Notice
New announcement from University of Rwanda. Check now!
5m
Keep Going 💪
You're doing great! Your app is running perfectly 🚀
6m
Keep Going 💪
You're doing great! Your app is running perfectly 🚀
⚠️ Urgent Update • 12m
Please check your app immediately. Impo...
Hello Vincent 👋 • 21m
Your FCM app is working perfectly

CamScanner